Problem Statement 1

ARC — Agentic Receivables Recovery and Section 138 Legal Action Pipeline

Domain: Trade receivables / channel finance / statutory legal recovery (BFSI-adjacent, manufacturing distribution)
Mandated stack: C# · .NET 9 · Microsoft Agent Framework 1.0 · Azure OpenAI (Microsoft Foundry) · Azure SQL · Azure Cosmos DB (NoSQL + integrated vector store) · Azure Functions (durable workflow host) · Azure Blob Storage · Azure AI Document Intelligence · Azure Service Bus · Microsoft Entra ID · Application Insights / OpenTelemetry
Nature of the build: Long-running, multi-day, human-gated, multi-agent workflow with hard statutory deadlines
Difficulty: High — this is the harder of the two problems. It is deliberately a *durability, determinism and governance* problem disguised as an AI problem.

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1. Business context

"PaintCo" is a large Indian decorative-paints manufacturer selling through ~2,900 appointed dealers, served by regional depots and covered by field sales officers (TSIs). Dealers buy on credit. When a dealer's account goes overdue, the company runs two interlocking processes.

Process A — the monthly ODOS (Outstanding Dues Over 90 days) demand-notice cycle

Day	Step	Owner
1st	System lists every dealer with ageing > 90 days **and** balance > ₹5,000	Portal
1st	Bulk email to all Depot Managers asking them to confirm notice generation, dealer by dealer	Portal
1st–7th	Approval window. **Default answer is Yes.** Manager may Accept or Decline; Decline requires a typed reason	Depot Manager
after 7th	Demand notices generated with the retained advocate's digital signature and despatched by HO Legal	HO Legal
after 7th	The same notice is surfaced to the covering TSI in the field mobile app	Field app
ongoing	TSI visits the dealer, records feedback and a recovery plan	TSI
monthly	The cycle repeats. Uncleared dealers receive the same letter again, indefinitely	—

Process B — Section 138 escalation (Negotiable Instruments Act, 1881)

Dealers lodge undated **security cheques** at appointment. If a demand notice produces no payment within 60 days, the company deposits the security cheque. If it bounces, criminal proceedings under Section 138 become available.

Step	Owner
System lists dealers who received a demand notice and paid nothing in 60 days	Portal
Depot Admin manually keys the cheque particulars — bank, cheque number, deposit date, deposit status — and uploads a scan	Depot Admin
Timeline freeze on the 15th	Portal
Realised → case closed. Bounced → proceed	Portal
Physical cheques couriered to Head Office; POD numbers keyed back into the portal	Depot Admin
HO receives the courier and **re-deposits every cheque at a single branch**	HO
Bounce/realisation status manually updated	Legal / PIS
Legal drafts the Section 138 notice and uploads it per dealer	Legal

The production snapshot that defines the problem

A single month's dashboard from the live system:

Metric	Value
Dealers with overdue > ₹5,000	2,526
Outstanding chased	**₹193.42 Cr**
Demand notices raised	2,526
Notices **approved** by Depot Managers	2,526
Notices **rejected**	**0**
Dealers **visited** by a TSI	**1**
Dealers not visited	2,525
Rows where ASM Remarks / HO Remarks are populated	**0**

The portal also carries an `Arrest Warrant` status column, which tells you exactly how far this escalation ladder goes.

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2. The problem statement

> **The system automates the production of legal paper and automates nothing about the recovery of money. It is open-loop: ₹193.42 Cr goes out as 2,526 notices and one dealer conversation comes back.**

Seven concrete failures follow from that, and the solution must address each of them explicitly.

P1 — Triggering is a single hard rule, not a decision. `ageing > 90 days AND balance > ₹5,000` fires everything. A ₹5,001 balance and a ₹3.3 lakh balance receive an identical letter with identical urgency. Nothing ranks dealers by recoverability, intent to pay, dispute probability, collateral strength or legal enforceability. The result is a permanent capacity mismatch: 2,526 notices pushed at a field force that can meaningfully service a few hundred dealers a month, so effort spreads uniformly across a distribution where value is heavily concentrated.

P2 — Approval is a timeout, not a judgement. 2,526 approved and 0 declined is what a rubber stamp looks like in data. This is not indiscipline. The Depot Manager is shown a dealer name and an amount and asked to authorise a legal action, with no view of *why* the balance exists. A gate with a permissive default and no decision support is an audit artefact, not a control.

P3 — "Outstanding" is never reconciled, so the claim amount may be wrong. The trigger reads gross open AR. In paint distribution a material share of apparent overdue is not default at all — unapplied credit notes, accrued scheme and rebate entitlements, damage and breakage claims, goods-return in transit, and cheques in clearing. Nothing nets these before a notice is raised. Demanding money a dealer does not owe is a legal exposure (wrongful notice, defamation, counter-claim), a relationship cost, and it silently corrupts every downstream metric. Escalating that same wrong number to a **criminal** complaint is materially worse.

P4 — The execution gap is the largest single value leak. 2,525 of 2,526 dealers were never visited. Notice generation is fully automated; the human follow-up that actually converts paper into cash is uninstrumented — not prioritised, not routed, not chased, not measured. With ASM and HO remark fields empty on every row there is no supervisory loop above the TSI either. The organisation currently cannot distinguish "*the dealer refused to pay*" from "*nobody asked.*"

P5 — Section 138 runs on a manual, physical, unguarded chain. Hand-keyed cheque particulars → courier → re-deposit at one branch → manual status updates → manual drafting. Every hop is a data-entry defect and an SLA risk, with no reconciliation between what the portal says the cheque did and what the bank's return memo actually says.

Worse: **Section 138 is statutorily time-boxed and nothing in the flow computes the clock.** The demand notice must issue within a fixed window of the dishonour intimation; the drawer then has a cure period; the complaint must be filed within a further window after that closes. A missed date does not delay the case — it extinguishes it. Today those dates live in an advocate's diary.

P6 — The context needed for any good decision is fragmented. Deciding whether to send one notice requires the ledger (SAP), dispute state (portal + email), the dealer agreement (PDF), the security cheque and its return memo (images), prior notices and their delivery proof, field history (free text in the app) and case status (with the advocate). Pointing an LLM at that raw sprawl produces exactly the predictable failure: hallucinated joins across systems, ambiguous metrics ("is ageing measured from invoice date or due date?"), and two different answers to the same question on two runs.

P7 — There is no learning loop. Uncleared dealers get the same letter every month forever. Nothing measures which lever actually recovered money — the notice, the visit, the Section 138 threat, or the cheque re-deposit. A dealer who has absorbed eleven identical notices without consequence has learned that the notice is noise, which is the precise opposite of its intent.

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3. Sub-problems to be solved

ID	Sub-problem	Success criterion
SP1	**Ledger reconciliation.** Net gross AR against unapplied credits, rebates, claims, returns and clearing items	A defensible `net_recoverable_exposure` per dealer with a full line-item audit trail
SP2	**Entity resolution / Dealer 360.** Bind one dealer across SAP code, portal ID, app ID, cheque account name and agreement party	"M/S Jai Bhavani Hardware" resolves to one entity, not four
SP3	**Recoverability scoring and prioritisation**	A ranked worklist; top-decile exposure identifiable in seconds
SP4	**Notice decisioning.** Replace default-Yes with an evidence-backed Issue / Hold / Reconcile recommendation	Every candidate carries a recommendation with cited evidence the manager can override with a reason
SP5	**Section 138 eligibility.** Validate enforceable debt, presentation validity and qualifying return reason *before* the physical chain starts	No ineligible case can enter the courier pipeline
SP6	**Limitation clock and drafting.** Compute every statutory date; draft from graph facts only	Zero statutory misses; zero model-generated amounts in a legal document
SP7	**Field orchestration and Promise-to-Pay.** Prioritised, geo-clustered visit plans; vernacular voice feedback captured as structured PTP and chased	Visit coverage on prioritised dealers; automated broken-PTP escalation
SP8	**Evidence and case-file assembly.** Ledger extract, invoices, cheque image, return memo, notice copy, POD, service proof	Court-ready bundle with a completeness score, gaps flagged before filing
SP9	**Supervision, learning and governance**	100% ASM/HO exception coverage; lever effectiveness measured per segment

Sequencing constraint that must be respected in the build: SP1–SP3 are upstream of everything else. No legal action is safe on an unreconciled number. The reconciliation agent must be demonstrable in shadow mode before any agent is allowed to influence a live notice.

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4. Mandated technical approach

4.1 Orchestration — Microsoft Agent Framework

Build the system as a **MAF `WorkflowBuilder` graph**, not as a loop of chat calls.

- **Two named workflows** sharing an upstream reconciliation stage:
- **Workflow A — ODOS cycle:** `A1 → A2 → A3 (gate: depot_manager_approval) → A5 (gate: advocate_signature) → A6`
- **Workflow B — Section 138 pipeline:** `A1 → A4 (gate: legal_progression) → A5 (gate: advocate_signature) → A7 (gate: legal_case_file_review)`
- **Human-in-the-loop is mandatory** at the four gates above, modelled with `RequestPort`. The graph must suspend, persist, and resume on the human decision event — including after a process restart.
- **Durability is non-negotiable.** A cycle spans 15+ calendar days (1st → 7th approval → 15th freeze). Use the **durable workflow host on Azure Functions** with an externalised checkpoint store so a run paused on the 1st resumes correctly on the 7th, on a different instance, with no session affinity.
- **Fan-out / fan-in** where genuinely parallel: risk scoring, document extraction and evidence collection across many dealers in one cycle. Use `AddFanOutEdge` / `AddFanInBarrierEdge`.
- **Conditional edges** for routing: a "visit" tier skips paper entirely; a `Reconcile` verdict terminates the branch and raises to Finance.
- **Idempotency:** every executor keyed on `(cycle_id, dealer_urn, node)`. Retries and checkpoint replays must not be able to double-issue a notice.
- **Tools as typed .NET functions** (`AIFunction` / `[Description]`-annotated methods), and expose the governed data surface as an **MCP server** so the same tool contracts serve agents, notebooks and future clients.
- **Middleware pipeline** for correlation IDs, PII redaction before the model call, token accounting and OpenTelemetry spans.

4.2 The deterministic / probabilistic boundary — the governing principle

This is the single most important design constraint and will be assessed directly.

Concern	Handled by
Money, dates, eligibility, statutory clocks	**Rules engine + SQL/graph queries. Never the LLM.**
Entity extraction from cheque scans, return memos, PODs	Azure AI Document Intelligence + vision model, schema-validated and confidence-gated
Summarisation, justification text, exception narratives	LLM
Planning, routing, tool selection	LLM (supervisor), bounded by the tool contract
Vernacular voice → structured Promise-to-Pay	Azure AI Speech + LLM, confirmed back to the TSI before commit

The LLM never computes a number that appears in a legal notice. It renders and explains numbers the deterministic layer produced. Any implementation where an amount, a date or an eligibility verdict originates in a model completion fails acceptance.

4.3 Knowledge and semantic layer

Implement in C# over Azure SQL + Cosmos DB.

Ontology (core classes): `Dealer` · `LegalEntity` · `Depot` · `Region` · `TSI` · `DepotManager` · `ASM` · `Invoice` · `CreditNote` · `Payment` · `LedgerPosition` · `AgeingBucket` · `SecurityCheque` · `ChequeReturnMemo` · `Collateral` · `DemandNotice` · `Section138Notice` · `CourierPOD` · `ServiceProof` · `FieldVisit` · `PromiseToPay` · `Dispute` · `LegalCase` · `Hearing` · `Advocate` · `Court`

Core relationships:

```
Dealer           —servedBy→       Depot   —inRegion→  Region
Dealer           —coveredBy→      TSI
CreditNote       —offsets→        Invoice
SecurityCheque   —dishonouredBy→  ChequeReturnMemo
DemandNotice     —precedes→       Section138Notice
Section138Notice —derivedFrom→    ChequeReturnMemo
FieldVisit       —yields→         PromiseToPay
Dispute          —blocks→         RecoveryAction
LegalCase        —supportedBy→    Evidence*
```

Physical-to-conceptual mapping is required so agents never reason over column codes — `ACB-X` → `DealerCategory` with credit-policy semantics; `S3` → `Region`; `B09` → `Depot`; `Notice Status = New` → `DemandNotice.lifecycleState = ISSUED_UNACKNOWLEDGED`.

Metric contracts — one definition, one owner, one version, resolved identically by the dashboard, the agent and the notice:

```
net_recoverable_exposure(dealer, asOf) =
      gross_open_AR
    − unapplied_credit_notes
    − accrued_scheme_rebates
    − goods_return_in_transit
    − cheques_in_clearing
    − disputed_under_review
```

Plus `odos_90` (pin ageing to **due date**, not document date — this must be explicit), `notice_eligible`, `s138_eligible`, `limitation_days_remaining`, `recoverability_score`.

Rules as versioned code artefacts, not prompt text:

ID	Rule
R1a	Hold the notice if the position is unreconciled
R1b	Hold the notice if `unapplied_credits / claim_amount > 0.40`
R1c	Hold if an open `Dispute` in `UNDER_REVIEW` exists, or an active `PromiseToPay` falls within grace
R2	`s138_eligible` requires a legally enforceable debt evidenced in the ledger, presentation within cheque validity, **and** a return-memo reason code in the qualifying set
R3	Statutory clock: notice-by date, cure-window close, file-by date. Alert at T-10, T-5, T-2 to a **named owner**, not a shared mailbox
R4	Segregation of duties — the agent that recommends can never approve
R5	Block all recovery action against a dealer under an insolvency moratorium
R6	No notice may quote an amount that lacks a resolvable lineage chain back to source rows

Rules must be independently unit-testable, versioned, and owned by Legal/Finance — changing a limitation period must be a signed-off config change, not a prompt edit.

Retrieval fuses three signals: graph traversal (what exists and how it connects), hybrid semantic search over notices/remarks/agreements/orders using the Cosmos DB vector store (what is related by meaning), and learned facts with confidence and provenance (what the system has discovered — "*Depot B09 posts credit notes with a systemic 9-day lag*", "*this dealer pays within 12 days of a visit but ignores letters*").

4.4 Required agent decomposition

ID	Agent	Core logic	Output	Gate
A1	Reconciliation	Deterministic netting per the metric contract	`net_recoverable_exposure` + line-item trail	auto
A2	Risk & Prioritisation	Scoring model on payment history + graph features + LLM read of qualitative TSI remarks	tier ∈	auto
A3	Notice Decisioning	Applies R1/R5; assembles evidence; drafts a one-line justification	Issue / Hold / Reconcile + citations	**Depot Manager**
A4	Legal Eligibility & Clock	**Rules engine decides, LLM only explains.** R2 + R3	eligible / blocked + `file_by_date` + alerts	**Legal**
A5	Drafting & Verification	Fills approved templates from graph facts only, then self-verifies entity, amount, cheque number, bank and dates field-by-field against the return memo	notice PDF for e-signature	**Advocate**
A6	Field Orchestration	Geo-clustered prioritised visit plans; ASR + LLM parse of vernacular voice into structured PTP; due-date chase	visit tasks, PTP records, broken-PTP escalations	auto + TSI confirm
A7	Evidence & Case File	Assembles and scores the bundle, flags gaps pre-filing	court-ready bundle + completeness score	**Legal**
A8	Supervisory Insight	NLQ over the semantic layer; auto-populates the empty ASM/HO exception fields; measures lever effectiveness	exception queue, analytics	auto

Cheque selection must be deterministic and explicit — newest bounced cheque that has a return memo, else newest cheque on file. Never left to the model.

4.5 Azure services map

Concern	Service
Models	Azure OpenAI in Microsoft Foundry — reasoning model for A3/A4 explanation, cheaper model for extraction and summarisation
Master data	Azure SQL (dealer, ledger, cheque, notice, case)
Conversation / workflow state / checkpoints	Azure Cosmos DB (NoSQL), partitioned by `cycle_id`
Vector store	Cosmos DB integrated vector search over policy, notices, orders, agreements
Documents	Azure Blob Storage, **WORM/immutable container** for legal artefacts, 8-year retention
Document AI	Azure AI Document Intelligence for cheque MICR, bank, amount, return-reason code
Voice	Azure AI Speech (Indian-English + regional locales)
Compute	Azure Functions (durable workflow host) + Azure Container Apps for the API
Messaging	Azure Service Bus for cycle fan-out and alert dispatch
Identity	Entra ID, managed identity end-to-end, no connection strings in config
Secrets	Azure Key Vault
Observability	Application Insights + OpenTelemetry traces from MAF middleware

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5. Edge cases the solution must handle explicitly

1. A1 cannot reconcile a position → mark `UNRECONCILED`, **exclude from notice eligibility**, raise to Finance. Never default to "assume owed."
2. The rules engine or clock service is unavailable → **fail closed**. No case proceeds.
3. A5 self-verification finds a mismatch between the draft and the return memo → block. No "best-effort" notice.
4. ASR confidence below threshold on a voice PTP → present the parsed record to the TSI for confirmation before commit; below a hard floor, discard.
5. The same dealer appears under two SAP codes with split exposure.
6. A credit note is posted *after* the notice is approved but *before* despatch.
7. Cheque image is illegible or the extracted MICR contradicts the hand-keyed portal entry.
8. Return memo reason code falls outside the qualifying set.

9. Dealer enters insolvency moratorium mid-cycle.
10. Depot Manager never responds — the gate must expire safely, and expiry must **not** mean "approved."
11. A checkpoint is restored on a different instance while a human decision is in flight.
12. Two cycles overlap for the same dealer.

6. Reference scenarios for the demo harness

Build with realistic synthetic data — ~2,500 dealers, 12 months of ledger history — and demonstrate at minimum:

#	Scenario	Expected behaviour
S1	Clean overdue, no credits, no dispute	Notice approved → drafted → visit task created
S2	77% of gross AR is an unapplied credit note	R1b blocks; verdict `Reconcile`; **no notice issued**; routed to Finance
S3	Genuine default with a bounced security cheque	Full path through to an assembled Section 138 case file
S4	Cheque bounced for a **non-qualifying** reason	R2 blocks before the courier chain starts
S5	Case with 3 days left on the filing clock	T-2 alert to a named owner; expiring-case escalation
S6	Dealer under moratorium	R5 blocks every action
S7	Vernacular voice PTP at 0.72 confidence	Presented to TSI for confirmation, not auto-committed
S8	Depot Manager approval arrives 6 days after pause	Workflow resumes correctly from checkpoint on a fresh instance
S9	Duplicate dealer identity across systems	Entity resolution merges before exposure is computed

7. Acceptance criteria

1. Both workflows are implemented as MAF graphs on .NET, with the four HITL gates modelled as `RequestPort` suspensions.
2. A workflow paused for a human decision survives a **full process restart** and resumes from checkpoint on a different instance.
3. Every amount, date and eligibility verdict in every generated artefact is traceable to a metric contract and source rows (R6). Demonstrate the lineage for at least one drafted notice.
4. No LLM output is used as the final authorisation decision anywhere in the system.
5. Rules R1–R6 are independently unit-tested, versioned, and changeable without touching prompts or agent code.
6. Section 138 statutory dates are computed by the rules engine and alerted at T-10/T-5/T-2; **limitation-miss rate is zero** on the test set.
7. Wrongful-notice rate below 1% of issued notices on the golden set; amount accuracy on drafted notices is 100% (any mismatch is a blocking defect).
8. Eligibility precision ≥ 0.95 and recall ≥ 0.98 against Legal-labelled cases.
9. Citation faithfulness ≥ 0.95 — every claim in an agent's justification resolves to a retrievable source.
10. RBAC and namespace isolation enforced server-side: a TSI-context call cannot read another region's dealers, and the check is not performed by the model.
11. Full OpenTelemetry trace per cycle, correlated end to end, with an immutable audit record of every gate decision (actor, role, verdict, reason, timestamp).
12. The system runs a complete cycle in **shadow mode** — computing everything, issuing nothing — as a demonstrable operating mode.

8. Deliverables

- Solution architecture and component design (C4-style or equivalent).
- MAF workflow topology diagram for both graphs, with gates and checkpoint boundaries marked.
- Agent and tool contracts (typed .NET signatures, MCP tool schemas).
- Ontology, metric contracts and the R1–R6 rule catalogue as versioned artefacts.
- Cosmos DB checkpoint/state and vector data models; Azure SQL master-data schema.
- Working C#/.NET 9 solution with synthetic data generator and a CLI or minimal API runner for scenarios S1–S9.
- Prompt and guardrail strategy, including the PII-redaction middleware.
- Evaluation harness: golden set (~200 labelled dealer-cases) and the metric report.

- Security, authorisation and audit design.
- Deployment (Bicep or Terraform) and operational runbook.

9. Stretch goals

- A2A protocol between the recovery workflow and an external advocate-side agent.
- Learning loop: measure lever effectiveness per segment and feed it back into A2's prioritisation.
- Autonomous action within policy bounds for the lowest-risk tier only, with a kill switch.
- Depot-manager override rate tracked as a system health metric — a depot with a zero override rate is itself an exception A8 should surface.

10. Assessment rubric

Dimension	Weight	What is being assessed
Deterministic/probabilistic separation	25%	Is the LLM anywhere on the critical path for money, dates or authorisation?
Durability and HITL correctness	20%	Does a multi-day, restart-surviving, resumable workflow actually work?
Semantic layer quality	15%	Ontology, metric contracts, lineage — or is it just RAG over PDFs?
Agent decomposition	15%	Are the boundaries principled, or is it one agent with eight prompts?
Security and auditability	15%	Server-side authorisation, immutable audit, RBAC, PII handling
Engineering quality	10%	Tests, idempotency, observability, deployability

A closing note on scope. Statutory periods under the Negotiable Instruments Act are stated here as the standard framework. In a real implementation they must be confirmed with counsel and held as versioned, Legal-owned configuration — precisely the kind of artefact that should never be baked into a prompt.

Appendix A — Technical Build Guide

> Code in this appendix is **illustrative scaffolding**, not copy-paste production code. Verify every API signature against the current Microsoft Agent Framework 1.0 documentation before you build — the framework reached GA recently and namespaces shifted during the RC period.

A.1 Solution structure

```
ARC.sln
├── src/
│   ├── ARC.Domain/                # Pure domain types. No Azure, no MAF, no LLM.
│   │   ├── Entities/              # Dealer, Invoice, CreditNote, SecurityCheque, ...
│   │   ├── Metrics/               # MetricContract, ExposureBreakdown, Lineage
│   │   ├── Rules/                  # IRule, RuleResult, R1a..R6 implementations
│   │   ├── Clocks/                # LimitationClock — statutory date arithmetic
│   │   └── State/                 # RecoveryState (the workflow state object)
│   ├── ARC.Data/                  # Repositories. Azure SQL + Cosmos DB.
│   │   ├── Sql/                   # EF Core / Dapper, read-only by default
│   │   ├── Cosmos/                # Conversation store, checkpoint store, vector store
│   │   └── Blob/                  # Document repository, SAS issuance
│   ├── ARC.Knowledge/             # Semantic layer
│   │   ├── Graph/                 # KnowledgeGraph — typed traversal over SQL/Cosmos
│   │   ├── Retrieval/             # Hybrid search + rerank + provenance
│   │   └── Extraction/            # Document Intelligence wrappers (cheque, memo, POD)
│   ├── ARC.Tools/                 # The governed tool surface
│   │   ├── ToolRegistry.cs        # AIFunction registration
│   │   └── Mcp/                   # Same tools exposed as an MCP server
│   ├── ARC.Agents/                # MAF executors + agent definitions
│   │   ├── Executors/             # A1..A8
│   │   ├── Workflows/             # OdosCycleWorkflow, Section138Workflow
│   │   └── Middleware/            # PII redaction, correlation, telemetry
│   ├── ARC.Host.Functions/        # Durable workflow host (Azure Functions)
│   ├── ARC.Api/                   # Minimal API for gates, dashboards, NLQ
│   └── ARC.Cli/                   # Scenario runner S1..S9
└── tests/
    ├── ARC.Domain.Tests/          # Rules + clock arithmetic. Highest-value tests.
    ├── ARC.Agents.Tests/          # Workflow topology, resume-from-checkpoint
    └── ARC.Eval/                  # Golden-set harness
```

Dependency rule to enforce: `ARC.Domain` references nothing. If a rule or a clock calculation needs an HTTP client, the design is wrong.

A.2 Package set

Package	Purpose
`Microsoft.Agents.AI`	Agent abstractions, `AIAgent`, threads
`Microsoft.Agents.AI.Workflows`	`WorkflowBuilder`, executors, edges, `RequestPort`, checkpointing
`Microsoft.Agents.AI.Workflows.Durable`	Durable host bindings for Azure Functions
`Microsoft.Extensions.AI`	`IChatClient` abstraction, `AIFunctionFactory`
`Azure.AI.OpenAI`	Azure OpenAI / Foundry client
`Azure.Identity`	`DefaultAzureCredential`, managed identity
`Microsoft.Azure.Cosmos`	State, checkpoints, vector search
`Microsoft.Data.SqlClient` / `Microsoft.EntityFrameworkCore.SqlServer`	Master data
`Azure.Storage.Blobs`	Document repository, user-delegation SAS
`Azure.AI.DocumentIntelligence`	Cheque and return-memo extraction
`Microsoft.CognitiveServices.Speech`	Voice PTP transcription
`ModelContextProtocol`	MCP server hosting
`Azure.Monitor.OpenTelemetry.Exporter`	Traces to Application Insights
`Polly`	Retry, circuit breaker on downstream systems

A.3 The state object

Everything the workflow knows travels in one serialisable object. Keep it flat, keep it versioned, and never put a live client or a `Stream` in it — it gets JSON-serialised into every checkpoint.

```
public sealed record RecoveryState
{
    public required string SchemaVersion { get; init; } = "1.0";
    public required string CycleId { get; init; }
    public required string DealerUrn { get; init; }      // resolved entity, not a SAP code
    public required DateOnly AsOf { get; init; }
    public required string CorrelationId { get; init; }

    public ExposureBreakdown? Exposure { get; init; }     // A1
    public RiskAssessment? Risk { get; init; }            // A2
    public NoticeVerdict? NoticeVerdict { get; init; }    // A3
    public EligibilityVerdict? Eligibility { get; init; } // A4
    public LimitationClock? Clocks { get; init; }         // A4
    public DraftArtifact? Draft { get; init; }            // A5
    public IReadOnlyList<VisitTask> VisitTasks { get; init; } = [];
    public IReadOnlyList<PromiseToPay> Promises { get; init; } = [];
    public CaseFile? CaseFile { get; init; }              // A7

    public IReadOnlyList<Approval> Approvals { get; init; } = [];
    public IReadOnlyList<ActionRecord> Actions { get; init; } = [];
    public string? TerminationReason { get; init; }
}

public sealed record ExposureBreakdown(
    decimal GrossOpenAr,
    decimal UnappliedCreditNotes,
    decimal AccruedSchemeRebates,
    decimal GoodsReturnInTransit,
    decimal ChequesInClearing,
    decimal DisputedUnderReview,
    decimal NetRecoverableExposure,
    IReadOnlyList<LineItemRef> LineItems,     // every component traceable
    bool IsFullyReconciled,
    string MetricContractVersion);

public sealed record LineItemRef(
    string SourceSystem,    // "SAP-FI-AR"
    string SourceTable,
    string SourceKey,       // document number, line
    decimal Amount,
    DateOnly PostedOn);     // <- this is what satisfies R6
```

Why `LineItemRef` matters: R6 says no notice may quote an amount without a resolvable lineage chain. That rule is only enforceable if lineage is a first-class field, not a log line. Candidates who model exposure as a `decimal` will fail acceptance criterion 3 and will not be able to fix it cheaply.

A.4 Executors and the workflow graph

An executor is a node. Keep the LLM inside the executor, never in the graph wiring.

```
public sealed class A1ReconciliationExecutor(IExposureService exposure)
    : ReflectingExecutor<A1ReconciliationExecutor>("A1_Reconciliation"),
      IMessageHandler<RecoveryState, RecoveryState>
{
    public async ValueTask<RecoveryState> HandleAsync(
        RecoveryState state, IWorkflowContext context, CancellationToken ct = default)
    {
        // Purely deterministic. No model call anywhere in this method.
        var breakdown = await exposure.ComputeAsync(state.DealerUrn, state.AsOf, ct);

        await context.AddEventAsync(new ExposureComputedEvent(state.DealerUrn, breakdown), ct);
        return state with { Exposure = breakdown };
    }
}
```

```
public sealed class A3NoticeDecisioningExecutor(IRuleEngine rules, AIAgent explainer)
    : ReflectingExecutor<A3NoticeDecisioningExecutor>("A3_NoticeDecisioning"),
      IMessageHandler<RecoveryState, RecoveryState>
{
    public async ValueTask<RecoveryState> HandleAsync(
        RecoveryState state, IWorkflowContext context, CancellationToken ct = default)
    {
        // 1. The rules decide. Deterministic, testable, versioned.
        var evaluation = rules.Evaluate(RuleSet.NoticeEligibility, state);

        var decision = evaluation.AnyBlocking
            ? NoticeDecision.Reconcile
            : NoticeDecision.Issue;

        // 2. The model only phrases the justification, and only from rule output.
        var rationale = await explainer.RunAsync(
            $"""
             Explain this notice decision to a Depot Manager in two sentences.
             Use only the facts below. Do not introduce any number not present here.

             DECISION: {decision}
             RULE VERDICTS:
             {evaluation.ToFactBlock()}
             """, cancellationToken: ct);

        return state with
        {
            NoticeVerdict = new NoticeVerdict(
                decision, evaluation.Verdicts, rationale.Text, evaluation.Citations)
        };
    }
}
```

Wiring the two graphs:

```
public static Workflow BuildOdosCycleWorkflow(IServiceProvider sp)
{
    var a1 = sp.GetRequiredService<A1ReconciliationExecutor>();
    var a2 = sp.GetRequiredService<A2RiskExecutor>();
    var a3 = sp.GetRequiredService<A3NoticeDecisioningExecutor>();
    var a5 = sp.GetRequiredService<A5DraftingExecutor>();
    var a6 = sp.GetRequiredService<A6FieldOrchestrationExecutor>();

    // Human gates
    var depotGate    = RequestPort.Create<NoticeApprovalRequest, NoticeApprovalResponse>("gate_depot_manager");
    var advocateGate = RequestPort.Create<SignatureRequest, SignatureResponse>("gate_advocate_signature");

    return new WorkflowBuilder(a1)
        .AddEdge(a1, a2)

        // Conditional: a 'visit' tier never generates paper at all.
        .AddEdge(a2, a6, condition: (RecoveryState s) => s.Risk?.Tier == RecoveryTier.Visit)
        .AddEdge(a2, a3, condition: (RecoveryState s) => s.Risk?.Tier is RecoveryTier.Notice or RecoveryTier.Section138)

        // Reconcile verdict terminates the branch — no notice, raise to Finance.
        .AddEdge(a3, depotGate, condition: (RecoveryState s) => s.NoticeVerdict?.Decision == NoticeDecision.Issue)

        .AddEdge(depotGate, a5)
        .AddEdge(a5, advocateGate)
        .AddEdge(advocateGate, a6)
        .WithOutputFrom(a3, a6)
        .Build();
}
```

Points the review will probe:

- The `Reconcile` verdict must *not* fall through to the gate. A blocked notice that still asks a human for approval reintroduces P2.
- The advocate gate must come *after* drafting and *after* self-verification, so the human signs a verified artefact, not a promise.
- `WithOutputFrom` needs both terminal nodes, because a terminated branch is a legitimate cycle outcome that must still be reported.

A.5 Human gates and the checkpoint store

The default in-memory checkpoint store is useless here — a cycle spans two weeks. Externalise it to Cosmos DB.

```
public sealed class CosmosCheckpointStore(Container container) : IJsonCheckpointStore
{
    public async ValueTask<string> SaveAsync(
        string runId, JsonElement payload, CancellationToken ct)
    {
        var doc = new CheckpointDocument
        {
            Id            = $"{runId}:{DateTimeOffset.UtcNow.Ticks}",
            RunId         = runId,
            CycleId       = ExtractCycleId(payload),
            Payload       = payload,
            CreatedUtc    = DateTimeOffset.UtcNow,
            SchemaVersion = "1.0"
        };
        await container.UpsertItemAsync(doc, new PartitionKey(doc.CycleId), cancellationToken: ct);
        return doc.Id;
    }

    public async ValueTask<JsonElement?> LoadAsync(
        string runId, string checkpointId, CancellationToken ct) { /* ... */ }
}
```

Running with resume:

```
var checkpointing = CheckpointManager.CreateJson(new CosmosCheckpointStore(container));

// Day 1 — run until the gate suspends it
var run = await InProcessExecution.StreamAsync(workflow, initialState, checkpointing);
string? pendingCheckpoint = null;

await foreach (var evt in run.WatchStreamAsync())
{
    switch (evt)
    {
        case RequestInfoEvent req:
            await approvalQueue.PublishAsync(req);      // email / portal / Teams card
            break;
        case SuperStepCompletedEvent step when step.CheckpointId is not null:
            pendingCheckpoint = step.CheckpointId;
            break;
    }
}
// Process exits here. Days pass.

// Day 7 — a different instance, cold start, approval has arrived
var resumed = await InProcessExecution.ResumeStreamAsync(
    workflow, pendingCheckpoint!, checkpointing);

await resumed.SendResponseAsync(requestId, new NoticeApprovalResponse(
    Approved: true, ActorUpn: "depot.manager@paintco.example", Reason: null));
```

The test that proves it (acceptance criterion 2): run the workflow to suspension, kill the process, start a fresh process with no shared memory, resume from the persisted checkpoint id, submit the approval, and assert the notice is drafted with the same exposure figure computed on day 1. If your test resumes in the same process, it proves nothing.

Gate expiry. Requirement: expiry must not mean approved. Model it as an explicit timer that submits a `NoticeApprovalResponse(Approved: false, Reason: "gate_expired")`. A silently-dropped gate is the same bug as default-Yes, wearing a different hat.

A.6 Rules engine

```
public interface IRule
{
    string Id { get; }              // "R1b"
    string Version { get; }         // "2026.03.1"
    RuleSet Set { get; }
    bool IsBlocking { get; }
    RuleResult Evaluate(RecoveryState state);
}

public sealed record RuleResult(
    string RuleId,
    string RuleVersion,
    RuleOutcome Outcome,            // Pass | Block | NotApplicable
    string Message,
    IReadOnlyList<Citation> Citations);
```

```
public sealed class R1b_UnappliedCreditRatio(decimal threshold = 0.40m) : IRule
{
    public string Id => "R1b";
    public string Version => "2026.03.1";
    public RuleSet Set => RuleSet.NoticeEligibility;
    public bool IsBlocking => true;

    public RuleResult Evaluate(RecoveryState state)
    {
        var e = state.Exposure
            ?? throw new InvalidOperationException("R1b requires A1 output. Fail closed.");

        if (e.GrossOpenAr <= 0)
            return new(Id, Version, RuleOutcome.NotApplicable, "No gross AR.", []);

        var ratio = e.UnappliedCreditNotes / e.GrossOpenAr;

        return ratio > threshold
            ? new(Id, Version, RuleOutcome.Block,
                  $"Unapplied credits are {ratio:P1} of gross AR (threshold {threshold:P0}). " +
                  $"Reconcile before any notice is issued.",
                  e.LineItems.Where(l => l.SourceTable == "BSEG_CREDIT").Select(Citation.From).ToList())
            : new(Id, Version, RuleOutcome.Pass, $"Unapplied credit ratio {ratio:P1} within tolerance.", []);
    }
}
```

Thresholds live in configuration, owned by Finance:

```
{
  "RuleConfiguration": {
    "Version": "2026.03.1",
    "ApprovedBy": "finance.controller@paintco.example",
    "ApprovedOn": "2026-03-14",
    "R1b": { "UnappliedCreditRatioThreshold": 0.40 },
    "R3":  { "NoticeWindowDays": 30, "CureWindowDays": 15, "FilingWindowDays": 30 },
    "R2":  { "QualifyingReturnCodes": ["FUNDS_INSUFFICIENT", "EXCEEDS_ARRANGEMENT",
                                       "ACCOUNT_CLOSED", "STOP_PAYMENT_QUALIFIED"] }
  }
}
```

Fail-closed discipline. Every rule throws if a prerequisite is missing rather than defaulting to `Pass`. A rule that can silently no-op when data is absent is worse than no rule, because it produces a clean-looking audit trail for an unchecked decision.

A.7 The limitation clock

This is the single highest-risk component. It gets its own service, its own tests, and its own availability target.

```
public sealed record LimitationClock(
    DateOnly DishonourIntimationDate,
    DateOnly NoticeByDate,
    DateOnly? NoticeServedDate,
    DateOnly? CureWindowEnds,
    DateOnly? FileByDate,
    int DaysRemaining,
    ClockStatus Status);            // Healthy | Warning | Critical | Expired

public interface ILimitationClockService
{
    LimitationClock Compute(ChequeReturnMemo memo, DemandNotice? notice, DateOnly asOf);
    IReadOnlyList<ClockAlert> DueAlerts(LimitationClock clock, DateOnly asOf); // T-10, T-5, T-2
}
```

Test cases that must exist:

Case	Assertion
Notice-by date lands on a Sunday or gazetted holiday	Behaviour is explicit and documented, not accidental
Memo received date differs from memo issue date	The clock uses the configured anchor and states which
Notice served date unknown	Cure window is `null`, status is `Warning`, not a guessed date
Cheque re-presented and bounced a second time	Which dishonour anchors the clock is an explicit policy decision
`asOf` after `FileByDate`	Status `Expired`; A4 blocks; escalation raised
Leap year and month-end arithmetic	30 days from 31-Jan resolves correctly
Timezone	All dates are `DateOnly` in IST; no `DateTime.Now` anywhere in the service

Never use `DateTime.Now` or `DateTime.UtcNow` inside domain code. Inject `TimeProvider`. Otherwise the clock is untestable and you cannot demonstrate acceptance criterion 6.

A.8 Azure SQL — core schema

```
CREATE TABLE dbo.Dealer (
    DealerUrn           VARCHAR(64)   NOT NULL PRIMARY KEY,   -- resolved identity
    SapCode             VARCHAR(20)   NULL,
    PortalId            VARCHAR(20)   NULL,
    AppId               VARCHAR(20)   NULL,
    LegalName           NVARCHAR(200) NOT NULL,
    Category            VARCHAR(20)   NOT NULL,   -- ACB-X, OTH  (map to DealerCategory)
    DepotCode           VARCHAR(10)   NOT NULL,
    RegionCode          VARCHAR(10)   NOT NULL,
    CoveringTsiId       VARCHAR(20)   NULL,
    MoratoriumFlag      BIT           NOT NULL DEFAULT 0,     -- drives R5
    CreditLimit         DECIMAL(18,2) NULL
);
CREATE INDEX IX_Dealer_Depot ON dbo.Dealer(DepotCode) INCLUDE (RegionCode);

CREATE TABLE dbo.LedgerPosition (
    PositionId          BIGINT IDENTITY PRIMARY KEY,
    DealerUrn           VARCHAR(64)   NOT NULL,
    DocumentType        VARCHAR(20)   NOT NULL,   -- INVOICE, CREDIT_NOTE, PAYMENT, REBATE, RETURN
    DocumentNumber      VARCHAR(30)   NOT NULL,
    DocumentDate        DATE          NOT NULL,
    DueDate             DATE          NULL,       -- ageing anchors HERE, not DocumentDate
    Amount              DECIMAL(18,2) NOT NULL,
    OpenAmount          DECIMAL(18,2) NOT NULL,
    ClearingStatus      VARCHAR(20)   NOT NULL,   -- OPEN, CLEARED, IN_CLEARING
    AppliedToDocument   VARCHAR(30)   NULL,       -- credit note -> invoice linkage
    SourceSystem        VARCHAR(20)   NOT NULL,
    SourceRowKey        VARCHAR(100)  NOT NULL,   -- lineage for R6
    CONSTRAINT FK_LedgerPosition_Dealer FOREIGN KEY (DealerUrn) REFERENCES dbo.Dealer(DealerUrn)
);
CREATE INDEX IX_Ledger_Dealer_Open ON dbo.LedgerPosition(DealerUrn, ClearingStatus) INCLUDE (DueDate, OpenAmount);

CREATE TABLE dbo.SecurityCheque (
    ChequeId            VARCHAR(40)   NOT NULL PRIMARY KEY,
    DealerUrn           VARCHAR(64)   NOT NULL,
    BankName            NVARCHAR(120) NOT NULL,
    ChequeNumber        VARCHAR(20)   NOT NULL,
    MicrLine            VARCHAR(40)   NULL,
    ChequeDate          DATE          NULL,
    Amount              DECIMAL(18,2) NULL,
    DepositedOn         DATE          NULL,
    Status              VARCHAR(20)   NOT NULL,   -- ON_FILE, DEPOSITED, BOUNCED, REALISED
    ImageBlobPath       VARCHAR(400)  NULL,
    ExtractionConfidence DECIMAL(4,3) NULL        -- from Document Intelligence
);

CREATE TABLE dbo.ChequeReturnMemo (
    MemoId              VARCHAR(40)   NOT NULL PRIMARY KEY,
    ChequeId            VARCHAR(40)   NOT NULL,
    ReturnReasonRaw     NVARCHAR(200) NOT NULL,
    ReturnReasonCode    VARCHAR(40)   NOT NULL,   -- mapped; drives R2
    MemoIssueDate       DATE          NOT NULL,
    MemoReceivedDate    DATE          NOT NULL,   -- clock anchor
    ImageBlobPath       VARCHAR(400)  NULL,
    ExtractionConfidence DECIMAL(4,3) NULL,
    CONSTRAINT FK_Memo_Cheque FOREIGN KEY (ChequeId) REFERENCES dbo.SecurityCheque(ChequeId)
);

CREATE TABLE dbo.GateDecision (          -- immutable audit of every human gate
    DecisionId          BIGINT IDENTITY PRIMARY KEY,
    CycleId             VARCHAR(40)   NOT NULL,
    DealerUrn           VARCHAR(64)   NOT NULL,
    GateId              VARCHAR(60)   NOT NULL,   -- gate_depot_manager, gate_advocate_signature
    ActorUpn            VARCHAR(200)  NOT NULL,
    ActorRole           VARCHAR(40)   NOT NULL,
    Decision            VARCHAR(20)   NOT NULL,   -- APPROVED, DECLINED, EXPIRED
    Reason              NVARCHAR(500) NULL,
    RecommendedAction   VARCHAR(20)   NOT NULL,   -- what the agent proposed
    WasOverride         AS (CASE WHEN Decision <> RecommendedAction THEN 1 ELSE 0 END) PERSISTED,
    DecidedUtc          DATETIME2     NOT NULL,
    CorrelationId       VARCHAR(64)   NOT NULL
);
```

`WasOverride` is a computed column on purpose — override rate is a headline system-health metric, and a depot with a permanent zero override rate is itself the exception A8 must surface.

A.9 Cosmos DB layout

Container	Partition key	Contents	TTL
`checkpoints`	`/cycleId`	MAF workflow checkpoints	90 days
`cycleState`	`/cycleId`	Per-dealer run state, terminal outcomes	none
`documents`	`/documentType`	Chunked policy, notices, agreements, orders + embeddings	none
`learnedFacts`	`/scope`	A-Box enrichment with confidence and expiry	365 days
`auditEvents`	`/cycleId`	Append-only agent/tool event log	8 years (archive tier)

Vector policy on `documents`:

```
{
  "vectorEmbeddingPolicy": {
    "vectorEmbeddings": [{
      "path": "/embedding",
      "dataType": "float32",
      "distanceFunction": "cosine",
      "dimensions": 3072
    }]
  },
  "indexingPolicy": {
    "vectorIndexes": [{ "path": "/embedding", "type": "diskANN" }],
    "includedPaths": [{ "path": "/documentType/?" }, { "path": "/effectiveFrom/?" },
                      { "path": "/status/?" }, { "path": "/regionScope/?" }]
  }
}
```

Filtered vector query — note the metadata predicates run *with* the similarity search, not after it:

```
SELECT TOP 8 c.id, c.documentType, c.version, c.chunkText, c.sourcePage,
       VectorDistance(c.embedding, @queryVector) AS score
FROM c
WHERE c.status = 'ACTIVE'
  AND c.documentCategory = @category
  AND (IS_NULL(c.regionScope) OR ARRAY_CONTAINS(c.regionScope, @region))
ORDER BY VectorDistance(c.embedding, @queryVector)
```

A.10 Tool contracts

```
public sealed class ExposureTools(IExposureService svc)
{
    [Description("Compute net recoverable exposure for a dealer as of a date. " +
                 "Returns the full deduction breakdown with source lineage. " +
                 "This is the ONLY authoritative source of a claim amount.")]
    public async Task<ExposureBreakdown> ComputeNetExposure(
        [Description("Resolved dealer URN, not a SAP code")] string dealerUrn,
        [Description("As-of date in yyyy-MM-dd")] string asOf)
        => await svc.ComputeAsync(dealerUrn, DateOnly.Parse(asOf), CancellationToken.None);
}

// Registration
var tools = new List<AITool>
{
    AIFunctionFactory.Create(exposureTools.ComputeNetExposure),
    AIFunctionFactory.Create(eligibilityTools.CheckSection138Eligibility),
    AIFunctionFactory.Create(clockTools.GetLimitationClock),
    AIFunctionFactory.Create(retrievalTools.SearchDocuments),
    AIFunctionFactory.Create(graphTools.TraverseGraph),
};
```

Tool-surface rule. There is no `SubmitNotice` tool available to any agent. Notices are emitted by the workflow after a gate, not by a model deciding to call a function. If a model *can* call it, a jailbroken model *will* call it.

A.11 Prompt patterns

Each agent's instructions follow the same shape. Keep them short — the guardrail lives in the tool surface and the rules engine, not in prose.

```
ROLE
You explain receivables decisions to a Depot Manager at a paints manufacturer.

HARD CONSTRAINTS
- Every number you state must appear verbatim in the FACTS block below.
- If a number you want to use is not in FACTS, say the figure is unavailable.
- You do not decide anything. The decision has already been made by the rules engine.
- Never suggest an action outside {Issue, Hold, Reconcile}.

STYLE
Two sentences. Plain business English. No hedging, no apology.

FACTS
{fact_block}
```

The `fact_block` is generated by code from typed objects, never by string-interpolating a database row. A candidate who builds it with `$"..."` over raw SQL output has reintroduced P6.

A.12 Document Intelligence extraction schema

```
public sealed record ChequeExtraction(
    string? BankName, decimal? BankNameConfidence,
    string? ChequeNumber, decimal? ChequeNumberConfidence,
    string? MicrLine, decimal? MicrConfidence,
    decimal? Amount, decimal? AmountConfidence,
    DateOnly? ChequeDate, decimal? DateConfidence,
    string? AccountName, decimal? AccountNameConfidence)
{
    public bool MeetsAutoAcceptThreshold =>
        ChequeNumberConfidence >= 0.90m && MicrConfidence >= 0.90m && AmountConfidence >= 0.85m;
}
```

Reconciliation rule: when extraction and the hand-keyed portal entry disagree on cheque number, MICR or amount, **neither wins automatically**. The case is flagged for Depot Admin correction and blocked from progression. This is the specific fix for the data-entry limb of P5.

A.13 Observability

Span naming convention — makes traces greppable across a 2,500-dealer cycle:

```
arc.cycle                       (root, per cycle)
  arc.dealer                    (per dealer run)
    arc.executor.A1_Reconciliation
      arc.tool.compute_net_exposure
      arc.sql.ledger_positions
    arc.executor.A3_NoticeDecisioning
      arc.rules.evaluate         { rule.set=NoticeEligibility, rule.blocking=R1b }
      arc.llm.explain            { model, prompt.tokens, completion.tokens }
    arc.gate.depot_manager       { suspended_utc, resumed_utc, wait_hours }
```

Required custom dimensions on every span: `cycle_id`, `dealer_urn`, `correlation_id`, `rule_version`, `metric_contract_version`, `prompt_version`.

Dashboards that must exist before go-live: gate wait-time distribution, override rate by depot, R1b block rate by region, clock status histogram (Healthy/Warning/Critical/Expired), self-verification failure rate on drafts.

A.14 Evaluation harness

```
public sealed record GoldenCase(
    string CaseId,
    string DealerUrn,
    DateOnly AsOf,
    decimal ExpectedNetExposure,
    NoticeDecision ExpectedNoticeDecision,
    bool ExpectedS138Eligible,
    DateOnly? ExpectedFileByDate,
    string LabelledBy,          // "finance" | "legal" | "both"
    string Rationale);
```

Metric	Computation	Target
Exposure accuracy	`abs(actual − expected) < ₹1`	100%
Notice-decision agreement	exact match on	≥ 0.92
S138 eligibility precision / recall	vs Legal label	≥ 0.95 / ≥ 0.98
File-by-date exactness	exact date match	100%
Wrongful-notice rate	notices issued where label = Reconcile	< 1%
Draft amount fidelity	drafted amount == metric contract output	100% (blocking)
Citation faithfulness	sampled claims resolving to a retrievable source	≥ 0.95
Limitation-miss rate	cases where `asOf > FileByDate` and status ≠ Expired	**0**

Run the harness in CI on every commit to `ARC.Domain`. Rules and clock arithmetic are the parts that must never regress silently.

A.15 Shadow mode

Implement as a first-class run mode, not a feature flag scattered through the code:

```
public enum RunMode { Shadow, Assisted, Live }
```

- **Shadow** — every agent runs, every artefact is generated and persisted, **no external side effect occurs**: no notice despatched, no visit task pushed, no leave/legal transaction committed. Outputs are diffed against what the legacy process did.
- **Assisted** — recommendations surface to humans; humans act.
- **Live** — approved actions execute.

Enforce it at the outbound adapter layer, not in agent logic. A single `IOutboundGate` that every side-effecting service must pass through, which throws in `Shadow`. That way a new agent added next year cannot accidentally bypass it.

A.16 Suggested build sequence

Week	Focus	Done when
1	Domain, metric contract, synthetic data generator (2,500 dealers, 12 months)	`ComputeNetExposure` returns a lineage-complete breakdown
2	Rules R1–R6 + limitation clock, fully unit-tested	Clock test matrix green, including holiday and re-presentation cases
3	A1/A2/A3 executors, ODOS workflow, in-memory checkpointing	Scenarios S1 and S2 run end to end
4	Cosmos checkpoint store + gate suspension/resume across process restart	Scenario S8 passes with a genuinely cold second process
5	Document Intelligence, A4, Section 138 workflow	Scenarios S3, S4, S5
6	A5 drafting with self-verification, A7 case file	Draft amount fidelity 100% on golden set
7	A6 field orchestration + voice PTP, A8 insight	Scenario S7 confirms below-threshold ASR to the TSI
8	Eval harness, telemetry, shadow mode, deployment	Full cycle in shadow with the metric report generated

A.17 Common failure modes in submissions

1. **One agent with eight prompts.** Reviewers will ask which component decides `s138_eligible`. If the answer is "the model, with a good prompt", the submission fails.
2. **Checkpointing in memory.** Demoed on a laptop in a single process, collapses on the first restart. Acceptance criterion 2 exists precisely to catch this.
3. **Exposure as a `decimal`.** Lineage bolted on later never quite works; R6 becomes unenforceable.
4. **Dates from the model.** `"Compute the filing deadline"` sent to an LLM. Confidently wrong, catastrophically so.
5. **RBAC in the prompt.** "*Only show documents for the user's region*" as an instruction rather than a query predicate.
6. **Gate expiry treated as approval.** Reimplements the exact problem the system exists to fix.
7. **No shadow mode.** Nothing can be safely validated before it touches a real dealer.
8. **Retrieval that ignores version and status.** Quoting a superseded policy in a legal notice.